Hail Identification Algorithm for DPR Onboard the GPM Satellite

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***Abstract*— Extreme precipitation, such as hail, has raised interest due to its high impact on human activities. In the new version of Global Precipitation Measurement (GPM) dual-frequency spaceborne weather radar (DPR) algorithm (version 7), a hail product is developed to identify hail along a vertical profile. The novelty of this algorithm offers the potential for retrieving a uniform and homogeneous hail dataset on the global scale from spaceborne radar sensor. The algorithm is built upon the precipitation-type index (PTI) developed for the GPM DPR. PTI has been shown to be effective in separating various precipitation types such as snow, graupel, and hail profiles. The capabilities of this algorithm to capture hail are validated by analyzing hail observations from various space and ground sources. These include ground validation radar Next Generation Radar (NEXRAD), GPM Microwave Imager (GMI)-based hail identification, and multiple scattering effect from Trigger module of DPR level-2 algorithm. The global-scale analysis demonstrates the good performance of the hail algorithm in identifying the worldwide high-frequency hail regions and seasonal transitions.**

***Index Terms*— Dual-frequency spaceborne weather radar (DPR), global distribution, Global Precipitation Measurement (GPM), hail identification, validation.**

I. Introduction

SCIENTISTS have dedicated great efforts to study hailstorms due to their destructive impact. An extreme hailstorm can cause billion-dollar losses in only several minutes. In 2021, there are 3763 major hail reports in the United States recorded by the database of the National Oceanic and Atmospheric Administration (NOAA) storm prediction center.

For years, the U.S. Next Generation Radar (NEXRAD) network and other high-resolution ground radars have played an important role in detecting and capturing hail observations. However, information from local ground radars is limited and hard to provide a global picture, not to say hail climatology. Satellite technique has provided new opportunities, bringing the potential of obtaining uniformity of hail measurements all over the globe. The Global Precipitation Measurement (GPM) constellation launched in 2014 carries the first dual-frequency spaceborne weather radar (DPR) onboard. The GPM-DPR operates at Ku-band (13.6 GHz) and Ka-band (35.5 GHz)

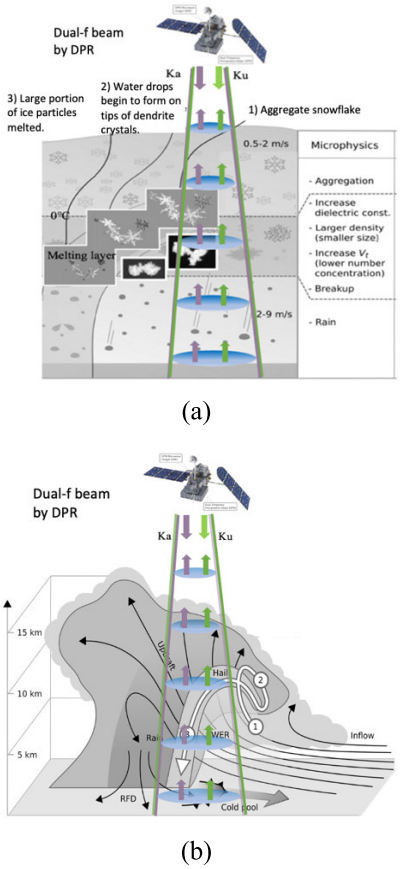


Fig. 1. Conceptual plot for GPM-DPR overlooking (a) stratiform storm featuring snow, melting layer, and rain on vertical profile, and (b) hailstorm on vertical profile.

improving our knowledge of precipitation processes relative to the single-frequency (Ku-band) radar used in Tropical Rainfall Measurement Mission (TRMM) by providing greater dynamic range, more detailed information on microphysics, and better accuracies in rainfall and liquid water content retrievals [1], [2], [3]. Fig. 1 shows a conceptual diagram for GPM-DPR observing various types of precipitation. Fig. 1(a) shows GPM-DPR overlooking stratiform precipitation featuring snow, melting layer, and rain along the vertical profile from top to bottom. In Fig. 1(b), a hailstorm with strong updrafts that support hail growth is captured by GPM-DPR.

Hail detection is relatively mature among high-resolution ground radar systems. Numerous methods to identify hail have been developed, some of which are in use with the NEXRAD network in the United States [4], [5], [6], [7]. Both single- and dual-polarization-based methods can detect hail occurrence,

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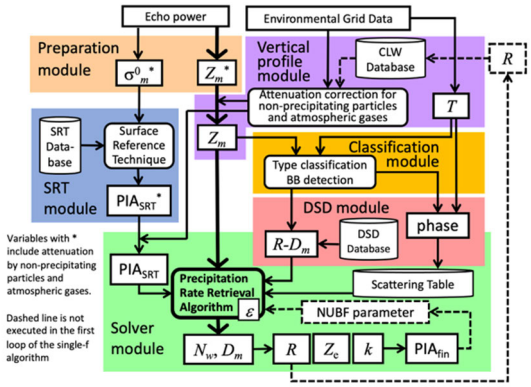


Fig. 2. DPR level-2 algorithm flow (GPM DPR ATBD [18]).

including its maximum sizes and shapes, especially with dual-polarized observation as differential reflectivity. Hail detection methods with space radar within tropical region using TRMM PR and radiometers are also well developed and documented in the literature [8], [9]. However, a spaceborne radar-based hail identification using dual-frequency radar of GPM-DPR is a relatively new effort. Seiki [10] studied the hail signals from DPR observations and heavy ice precipitation detection approach was described in [11]. These are some of the pioneer works on the study of DPR observations for hail detection.

The spaceborne hail retrievals provide an opportunity for uniform detection and global coverage of hail estimates that ground reports and ground-based radar networks alone cannot supply. Spaceborne radar only provides reflectivity information at each resolution volume, lacking dual-polarized, and Doppler information. The resolution volume is much coarser than ground radar with 125 m in the vertical direction and widens in the horizontal direction when the beam reaches toward surface (around 5 km near surface). However, different from traditional ground radar, DPR has two frequency channels well aligned with each other. The vertical profile of Ku-band and Ka-band observations from DPR and their differences [also called measured dual-frequency ratio (DFRm)] inlaid rich information of microphysics, including precipitation types and melting status.

We have been engaged in developing various algorithms for DPR in the dual-frequency classification module using DFRm and reflectivity [12]. Stratiform and convective rain types are classified with the features on DFRm [13]. Melting layer heights along the vertical profile are detected with characters of DFR derivatives [14]. The precipitation-type index (PTI) extracts valuable knowledge on both reflectivity profile and its ratio and has been applied to identify hydrometeor types such as snowfall and graupel [15], [16], [17]. Carrying these legacies, we develop a new method to perform hail identification along each vertical profile for GPM-DPR. A new product of "flagHail" has been implemented in version 7 of DPR level-2 algorithm and has been public since December of 2021. This flag outputs either "1" or "0" indicating whether hail exists along the profile and is one of the products implemented in the classification module from the DPR level-2 algorithm, as shown in Fig. 2.

PTI plays an important role in a couple of algorithms in profile classification. PTI is a value calculated for each dual-frequency profile with precipitation observed by GPM DPR. The DFR slope with respect to height, the maximum of reflectivity, and the storm top height are three key ingredients composing PTI value. PTI is effective in separating snow and graupel/hail (GH; either graupel or hail exists) profiles. In [16] and [17], 85% of snow profiles have PTI values larger than 17 and 85% of GH profiles are with PTI less than 5.5. Here, we zoom in further on PTI for GH profiles and separate graupel and hail profiles with different PTI thresholds. A new Boolean product of "flagHail" is a hail-only identifier for each vertical profile. This hail product is validated with ground radar hail products, other DPR products from Trigger module of DPR level-2 algorithm, and radiometer-based hail identification algorithm.

Section II describes the vertical feature for hail profile. Algorithm development is illustrated in Section III. The algorithm is validated for cases observed synthetically by GPM DPR and the NEXRAD network as well as with a GMI-based approach. These analysis activities are illustrated in Section IV. A global map is generated and compared with other hail products, studying the global distribution and seasonal transitions in Section V. The summary and conclusions are drawn in Section VI.

II. Vertical Profile Features for Hail Precipitation

A vertical description of the profiles of precipitation is a long-term goal of atmospheric research and precipitation science [19]. In a typical stratiform rain profile, spaceborne radar echoes illustrate clear boundaries of solid, melting, and liquid phases of precipitation from top to bottom. However, for convective precipitation such as convective rain, graupel, or hail, the transition of phases is relatively vague, but vertical profiles have their own features.

A. Background on Dual-Frequency Vertical Profile

GPM-DPR offers two independent observations at Ku- and Ka-bands. The difference of two observations is called DFRm. DFRm (in dB scale) is defined as

$$\mathrm{DFR}_m = 10\log_{10}(Z_m(K_u)) - 10\log_{10}(Z_m(K_a)) \tag{1}$$

where Z_m is the measured equivalent radar reflectivity factor in linear scale. Z_m can be related to equivalent radar reflectivity factor Z_e through

$$Z_m = Z_e \cdot A = Z_e \exp\left[-0.2\ln(10)\int_0^r k(s)ds\right] \tag{2}$$

where A is the cumulative attenuation factor from radar to the bin of interest. Z_e is related to the particle size distribution $N(D)$ and the backscatter cross section σ_b of the hydrometeors for a given wavelength λ. Taking log of both sides, (2) can be expressed as

$$\begin{aligned} &10\log_{10}(Z_m) \\ &\quad = 10\log_{10}(Z_e) - \log_{10}\left(\exp\left[0.2\ln(10)\int_0^r k(s)ds\right]\right) \\ &\quad = 10\log_{10}(Z_e) - \mathrm{PIA} \end{aligned} \tag{3}$$

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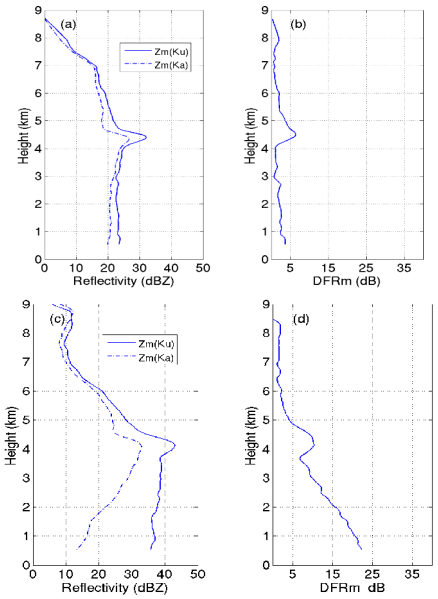


Fig. 3. Typical hail-free vertical profile for (a) and (b) stratiform and (c) and (d) convective rain from airborne radar data. (a) and (c) Reflectivity measurement at Ku- and Ka-bands and (b) and (d) DFRm.

where

$$\mathrm{PIA} = -10\log_{10}(A). \tag{4}$$

PIA (in dB) is a positive number and denotes the two-way attenuation from radar to the range of interest. Radar DFR in dB, describing the difference of reflectivity factor between two frequency channels, is defined as

$$\mathrm{DFR} = 10\log_{10}(Z_e(K_u)) - 10\log_{10}(Z_e(K_a)). \tag{5}$$

Substituting (5) in (1) results

$$\begin{aligned}\mathrm{DFR}_m &= 10\log_{10}(Z_m(K_u)) - 10\log_{10}(Z_m(K_a)) \\ &= \mathrm{DFR} + \delta\mathrm{PIA}\end{aligned} \tag{6}$$

where δPIA is the attenuation difference between Ku- and Ka-bands expressed in a decibel scale.

DFRm is a unique observation from GPM-DPR, and it inlaid richer hydro-type information than the reflectivity profile. In (6), DFRm is composed of DFR and δPIA. DFR is controlled by the non-Rayleigh scattering of precipitation particles, while δPIA counts for the two-way attenuation difference from radar to the range of interest. When the radar beam from the GPM core satellite propagates through precipitation, such as the melting layer or moderate to heavy rain, it undergoes large attenuation, and normally, Ka-band attenuation is greater than Ku-band. Meanwhile, both Ku- and Ka-band reflectivity is affected by non-Rayleigh scattering effect, with the Ka-band being more severe than the Ku-band. This makes the DFRm a viable parameter for making inferences about hydrometeor. The vertical features of the DFRm profile have been studied for different precipitation types. In the classification module of DPR level-2 algorithm, the DFRm is a key parameter that is widely used in the dual-frequency classification of vertical profile.

Fig. 3 shows a typical hail-free vertical profile of reflectivity and DFRm for stratiform and convective rain from airborne radar observation. Fig. 3(a) shows the Ku- and Ka-band reflectivity profile with typical bright band feature for a stratiform rain. The corresponding DFRm profile is shown in Fig. 3(b). At altitude above about 5 km, DFRm values are small due to the small particle size of ice and increase slightly with the decrease of height. The particles start to aggregate when they fall, which explains the slight increase in the value with a decrease in the height. In the ice region, the attenuation difference δPIA is negligible compared with DFR, and thus, the shape of DFRm is controlled by DFR. Below 5 km, the DFRm increases sharply until it hits a local maximum, and then, it decreases with decreasing height until it reaches a local minimum value. The rapid change in the shape of DFRm is associated with the precipitation process in the melting layer, where it is controlled by both DFR and δPIA. From around 4 km and below, the DFRm profile continues to increase, and this is the liquid phase. In this region, the DFR values do not change much due to balance in the various precipitation processes. The increase in the DFRm value is mainly contributed from the attenuation difference. Fig. 3(c) and (d) shows a case for convective rain profile. Reflectivity values are larger at both frequencies, but the DFRm profile shares similar features as shown for stratiform rain in Fig. 3(b) due to the phase transition of hydrometeors. The increase of the DFRm value in rain is obvious for both stratiform and convective rain, but the slope of DFRm with respect to the height is much larger for convective than for stratiform rain.

B. Dual-Frequency Vertical Profile With Hail Contamination

The features of the hail-contaminated profiles from DPR observation differ from those hail-free profiles shown in Fig. 3. We collected around 200 DPR profiles from hailstorms validated by the NEXRAD radar network from years 2015 to 2017. Only profiles with peak reflectivity of Ku-band exceeding 45 dBZ are chosen. Fig. 4 shows the medium profile for this dataset with Ku-band, Ka-band, and DFRm being blue, red, and yellow, respectively. From Fig. 4(a) to (c), the hail profiles are grouped with increasing reflectivity peak value at the Ku-band, which indicates an increasing hail contamination level. One obvious feature is that the peak of the Ka-band profile occurs higher than the Ku-band for hail-contaminated profile. These two peak locations align well for hail-free profile in Fig. 3(a) and (c). In Fig. 4, the slope of Ku-band increases sharply with decreasing height before it hits a maximum due to the water-phase transition. However, the slope of Ka profile does not show the same sharp increase as in Ku. The nontypical profile shape of Ka-band is mainly caused by the joint contributions from strong attenuation above 0° isotherm due to the heavy lift of water and the multiple scattering effects. Below the peak position for Ku profile, the slope of reflectivity decreases obviously due to attenuations from melting hail or heavy rain. This slope for the Ka-band is, however, less than Ku, breaking the single scattering rules, indicating that multiple scattering enhancement took place.

The yellow profile in Fig. 4 is the DFRm profile with hail. The DFRm peak value for hail profile can be as large

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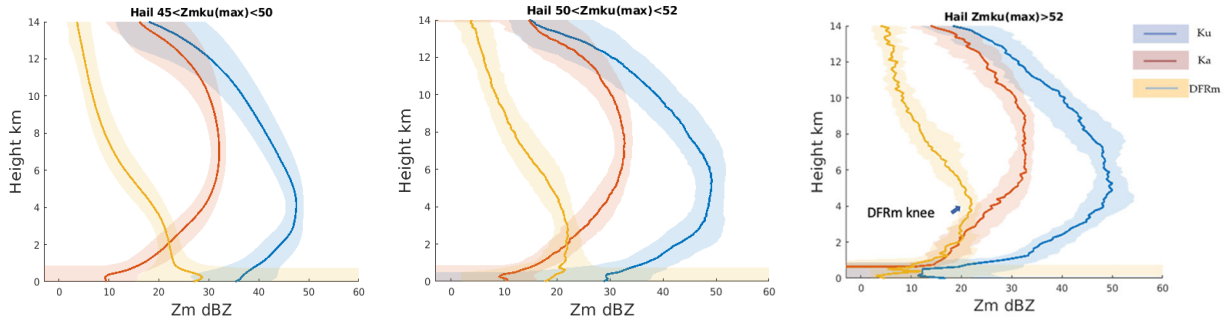


Fig. 4. Typical vertical profile by GPM-DPR with various hail contamination. (a)–(c) Hail profiles with increasing peak value of reflectivity at Ku-band. Solid lines are the medium profiles of reflectivity with Ku-band, Ka-band, and DFRm being blue, red, and yellow, respectively. The semitransparent envelope marks the region between the 25th and 75th percentiles.

as 20 dB or even beyond. Convective rain profile, although the Ku peak value can also be large, the DFRm peak is normally less than 15 dB. This large DFRm peak is due to a large Ku peak value but a relatively smaller Ka peak due to strong attenuation. After reaching a peak value, the DFRm profile remains relatively constant or even decreases slightly due to various levels of hail contamination. This behavior is quite different from features of hail-free profiles. For both stratiform and convective rain profile, there is a clear DFRm local minimum below the 0° isotherm, which is an indication of transition from mixed phase to liquid. Below that, DFRm values increase monotonically because of attenuation difference increase in the liquid phase. However, for hail profiles, attenuation difference does not increase this obvious or even decrease sometimes below 0° isotherm due to a result of a complex mixed effect from non-Rayleigh scattering, multiple scattering, and attenuations. Therefore, from Fig. 4(a) to (c), the DFRm profile changes from a more "convective-like" profile to the profile with clear a "knee" shape indicating strong hailstorms. The characteristics described above are useful features for developing a hail detection algorithm for DPR radar.

III. Algorithm Development

A. Precipitation-Type Index (PTI)

In the dual-frequency classification module of GPM-DPR level-2 algorithm, DFRm and its vertical features have been widely used to perform hydrometeor-type classification. A PTI was developed to effectively classify snow profile and GH profile in our team's previous work (product of "flagSurfaceSnowfall" and "flagGraupelHail") [15], [17]. PTI is defined here in the following:

$$\mathrm{PTI} = \frac{\mathrm{mean}(\mathrm{abs}(\mathrm{DFR}m_{\mathrm{slope}}))}{\mathrm{Zmku}_{\mathrm{max}} * \mathrm{Storm_top_height}}. \tag{7}$$

In (7), the DFRm slope (in dB/km) is along the vertical height. The average of the DFRm slope is taken from storm top to clutter-free region. $\mathrm{Zmku}_{\mathrm{max}}$ (in dBZ) represents the maximum value of reflectivity at Ku-band. Storm_top_height represents the altitude of storm top in km and is included in PTI to further enlarge the separation of different hydrometeor types.

Fig. 5(a) shows the histogram plot of PTI for snow and GH profiles. Clear separation of PTI value range can be seen

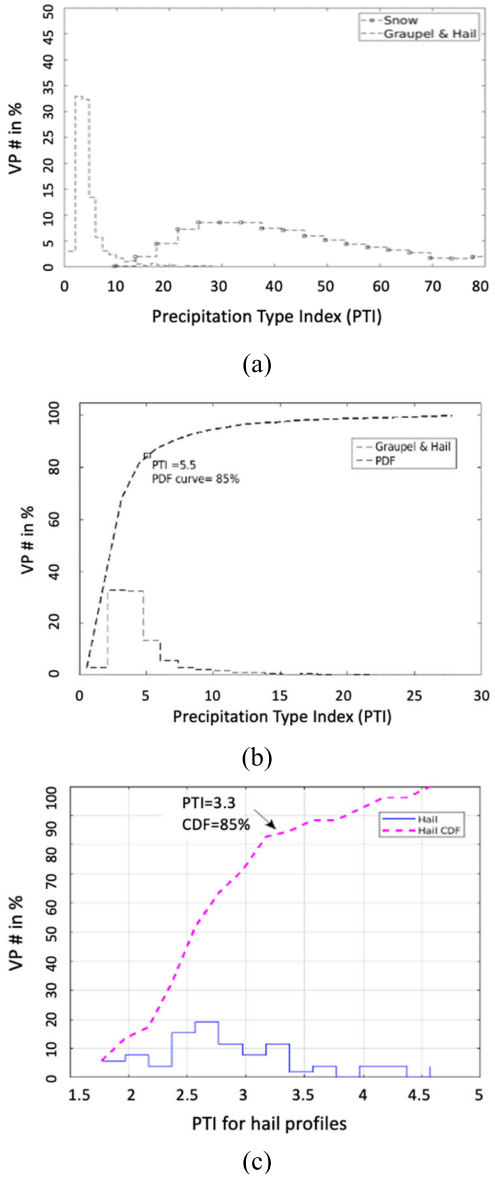


Fig. 5. (a) Histogram of PTI for snow and GH profiles. (b) Enlarged histogram plot for GH profiles. (c) Histogram of PTI for hail-only profiles.

from Fig. 5(a) for snow (dashed line with circle) and GH (dashed line) profiles. Snow profiles have PTI values almost exclusively larger than 10 based on more than 4000 snow profiles data. However, PTI for GH profiles falls on small end,

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TABLE I
QUALITATIVE RANGE OF THREE COMPONENTS IN PTI FOR STRATIFORM RAIN, SNOW, GH, AND HAIL

	Stratiform rain	Snow	GH	Hail
DFRm slope	small	large	medium	Medium to small
Z max	small	small	large	larger
STH	small	small	large	larger
PTI	medium	large	small	smaller

TABLE II
LIST OF INDEPENDENT HAIL VALIDATION CASE

Hail Validation Cases			
GPM DPR orbit #	Date	Validation sources	Geolocation
8630	09/05/2015	Marra et al. 2017. Hail report and GR	city of Naples in Italy
519	04/02/2014	NEXRAD radar KFWS	Dallas Fort Worth, USA
12594	05/17/2016	Mroz et al. 2018. DPR NUBF features, GR	Offshore Miami, USA
1518	06/05/2014	Mroz et al. 2017. NEXRAD radar KGLD	Border of Colorado and Kansas, USA
30581	07/17/2019	Battaglia et al. 2022. GR, GMI.	South Dakota, USA
37918	10/31/2020	Battaglia et al. 2022. "Champion storm" GMI.	Tasman Sea, east of Sydney, Australia
33877	02/14/2020	Battaglia et al. 2022. "Champion storm"	Northern Argentina
6600	04/27/2015	Battaglia et al. 2022. "Champion storm"	Texas, USA
25028	07/25/2018	NEXRAD radar KLNX	North Platte, Nebraska, USA
25427	08/20/2018	NEXRAD radar KLZK	Little Rock, Arkansas, USA
6906	05/11/2015	NEXRAD radar KFWS, Trigger module	Dallas Fort Worth, USA
24721	07/05/2018	Laviola et al. 2020. GMI	Veneto region in Italy
24557	6/25/2018	NEXRAD radar KVNX	Vance AFB, OK
24198	6/2/2018	NEXRAD radar KFSD	Sioux Falls, SD

with 85% of profiles having PTI less than 5.5. Fig. 5(b) shows an enlarged histogram plot for GH profile for clear illustration. In general, the absolute value of DFRm slope in the numerator of (7) is large for a profile with snow particles due to the fast increase of size during aggregation falling toward the surface. Normally, large reflectivity and DFRm values can be observed associated with large-rimed GH particles and large attenuation. However, the slope of the DFRm for GH is smaller than aggregates. Since in their formation, the heavily rimed particles are lifted by the updraft, there is no strong indication of one-way size increase or attenuation difference increase. The peak of Ku-band reflectivity in the denominator of (7) is larger for GH than for snow, while storm top height in the denominator is usually lower for snow and stratiform rain than for profile with GH [20].

Besides GH profile (either graupel or hail exists), hail-only profiles are the focus of this study. Fig. 5(c) shows PTI for hail-only profiles, which are validated with ground dual-polarization radar observations. The histogram of hail PTI falls in an even smaller value than GH profiles in Fig. 5(b); 85% of hail profiles have PTI smaller than 3.3. A large part of it is due to the multiple scattering features

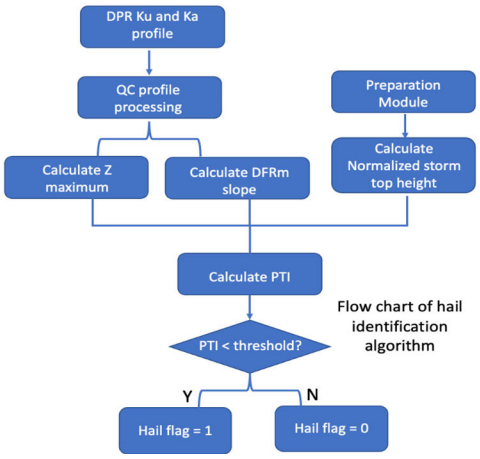


Fig. 6. Flowchart of hail identification algorithm.

associated with hail profiles that decrease the slope of DFRm. Meanwhile, storm top height and maximum of reflectivity for hail profiles are in a higher value range for hailstorms, thus further decreasing the PTI. Based on this concept, we develop a hail-only detection algorithm for DPR on a profile basis. Fig. 6 shows the flowchart of the algorithm. This algorithm has been implemented in the classification module of version 7 of DPR level-2 algorithm (DPR level-2 ATBD [18]). Table I lists the qualitative range of the three components in PTI for snow, stratiform rain, GH, and hail.

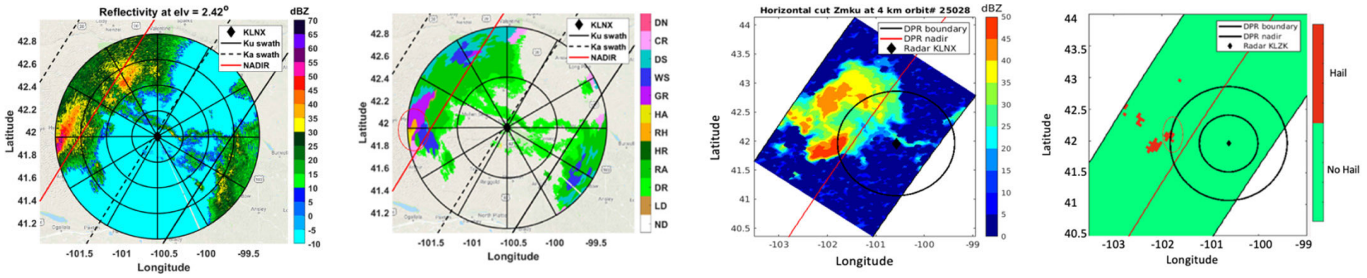
Originally, the GPM stakeholder community wanted to know the presence of graupel and hail, and therefore, a "Graupel or Hail" (high-density ice) product was developed. However, over time, the interest in hail increased due to societal impact and a separate hail product was developed to satisfy that need. Products of "flagGraupelHail" and "flagHail" are two different products in the DPR level-2 algorithm with the previous one stored in the experimental module. The focus of this article is on the hail identification algorithm embedded in the product of "flagHail" in the classification module of GPM DPR level-2 algorithm.

IV. ALGORITHM EVALUATION AND VALIDATION ON HAILSTORM

Validation is crucial for any algorithm development. The hail identification algorithm has been validated not only from independent hailstorms but also from the global hail distribution and trends. Case-based validation sources include hail report, ground radar network, radiometer-based retrieval, and multiple scattering product from the Trigger module [21]. Table II lists the hail cases we have been successfully validated with. These independent hail cases are confirmed from various reference sources ranging from 2014 to 2020 [22], [23], [24], [25], [26]. In Section IV, we will illustrate several sample validation comparisons in detail.

The global hail distribution map is generated and cross-compared with hail maps generated using space radar and radiometer-based algorithms by Mroz et al. [24] and Bang and Cecil [27].

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(a) (b) (c) (d)

Fig. 7. Precipitation event observed simultaneously by KLNX radar located in North Platte, NE, and GPM DPR with orbit #25028 on 07-25-2018 around UTC 11:25:49. (a) KLNX PPI scan at elevation of 2.42°. (b) Hydrometeor type for scan in (a). (c) Ku-band reflectivity at 4 km from DPR. (d) Hail flag for (c).

A. Validation With NEXRAD Radar

The NEXRAD is a network of 160 high-resolution S-band Doppler dual-polarimetric weather radars operating in the United States. Hailstorms identified by the NEXRAD radar network are selected first due to the sophisticated hydrometeor-type classification algorithms developed for this dual-polarization radar system [28], [29]. Then, we search for GPM satellite data that capture the same storm within minutes to make sure that both radars observe at the same hail event.

The validation case shown in Fig. 7 is a strong storm simultaneously observed by DPR (orbit #25028) and KLNX radar located in North Platte, NE on 07-25-2018 around UTC 11:25:49. Fig. 7(a) shows the plan position indicator (PPI) scan of reflectivity at 2.42° elevation angle for KLNX. The black circles are 33-, 66-, and 100-km range of KLNX radar. Red, black dashed, and black solid lines represent the DPR nadir, inner swath, and DPR full swath, respectively. Fig. 7(b) shows the hydrometeor type for scan in Fig. 7(a) using the algorithm described in [28]. To the west of radar at about 100-km range, hail is detected, illustrated with yellow color in Fig. 7(b). DPR Ku reflectivity at 4 km is shown in Fig. 7(c). The black circle corresponds to a 100-km range of KLNX. In Fig. 7(d), the hail flag is illustrated with red stands for hail locations. Comparing Fig. 7(b) and (d), at DPR and KLNX overlapped region, we detect the hail at the west to the KLNX radar, the same spot in Fig. 7(b).

Another sample validation case shown here is with NEXRAD radar KLZK located at Little Rock, AR. A strong storm was simultaneously observed by DPR (orbit #25427) and KLZK radar on 08-20-2018. Fig. 8(a) shows the PPI scan of reflectivity at 3.12° elevation angle for KLZK. The black circles are 25-, 50-, 75-, and 100-km range of KLZK radar. Fig. 8(b) and (c) shows the hydrometeor type for the scan at elevation of 3.12° and 4°, respectively. Circled areas have rain and hail on the northwest side of KLZK. DPR Ku reflectivity at 4 km is shown in Fig. 8(c). The black circle corresponds to the 100-km range of KLNX. Fig. 8(e) shows the hail flag with red stands for hail locations found at northwest of KLZK. Comparing Fig. 8(b), (c), and (e) from both DPR and KLZK overlapped regions, we detect the hail at the same location precisely.

Mroz et al. [24] analyzed a thunderstorm captured by the NEXRAD radar of KGLD and GPM DPR orbit #1518 on 06-05-2014. Fig. 9(a) and (b) shows the horizontal cut of Ku- and Ka-band reflectivity at 4 km height. The core of the storm features a high value of reflectivity above 50 dBZ. The huge contrast between Ku- and Ka-band reflectivity values indicates that strong attenuations reach around 20 dB from storm top downward to 4 km height. Hail is detected as shown in Fig. 9(c) using the hail identification algorithm described in Section III. The nearby KGLD radar captured the same storm, and the dual-polarization parameters were gridded into DPR resolution. Fuzzy-logic hydrometeor classification (HID) using Dolan et al. [29] algorithm was applied to the gridded data. Then 3-D HID products are projected to a 2-D surface within DPR point of view. This projection of HID is shown in Fig. 10(d). Fig. 9(d) is the same as in [24, Fig. 3]. The red color represents "Hail" in the plot. Since the hail identification algorithm detects hail along the profile, it is a precise comparison for 2-D projected HID in Fig. 9(d).

Since we regard HID from ground radar as "ground truth," the similar pattern of hail in Fig. 9(c) and (d) demonstrates the excellent performance of hail identification algorithm. Fig. 9(e) shows the vertical cut of Ku-band reflectivity at storm center (scan #4996), where storm structure can be clearly observed. Large reflectivity values above 50 dBZ extend as high as 14 km, the typical feature of hailstorms. Fig. 9(f) and (g) shows selected vertical profiles of Ku- and Ka-band reflectivity and DFRm. For both profiles, Ku- and Ka-band reflectivity peaks are above 45 and 30 dBZ above freezing level. DFRm knee appears at around 5 km. These are common hail profile characteristics discussed in Section II.

B. Validation With GMI

There is pioneering research focusing on the relationships between hail occurrence and the signal of high-frequency microwave channels [25]. The frequency of 19, 37, and 85 GHz of TRMM Microwave Imager (TMI) onboard the TRMM satellite was used to detect hail at the surface. Using 16+ years of data from the TRMM, Cecil and Blankenship [9] paired TRMM brightness temperature-derived precipitation features with surface hail reports in the United States to train hail retrieval on passive microwave data. Ever since the launch in 2014, the potential offered by the GPM constellation for monitoring severe storms is unprecedented [2]. Bang and Cecil [27] applied this hail retrieval to features in the GPM to develop a nearly global passive microwave-based climatology of hail. Laviola et al. [26] proposed a method for detecting hailstorms using a probability-based model originally designed for Advanced Microwave Sounding

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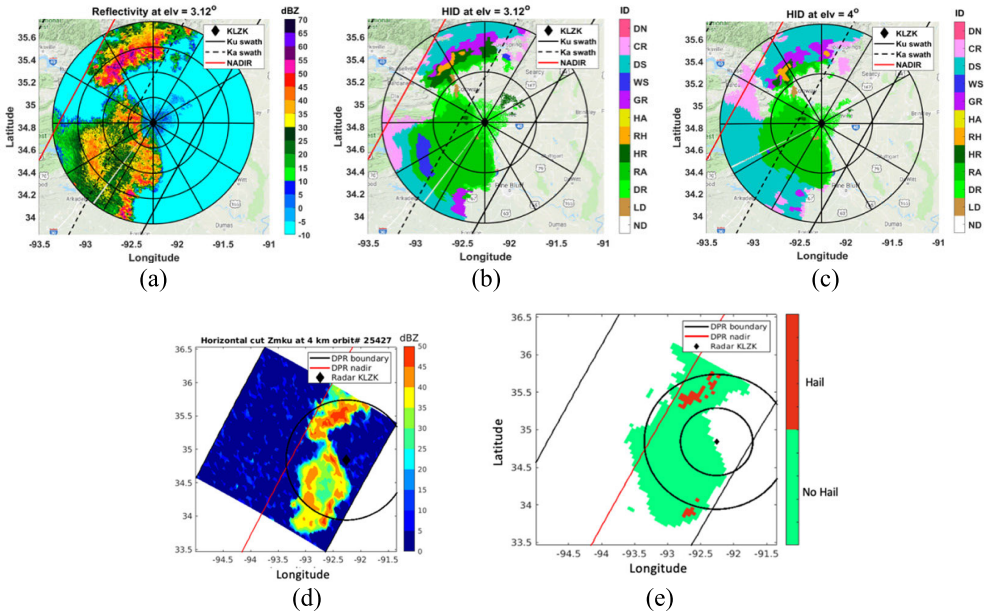


Fig. 8. Precipitation event simultaneously observed by KLZK radar located at Little Rock, AR, and GPM DPR with orbit #25427 on 08-20-2018. (a) PPI reflectivity at 3.12° elevation angle for KLZK. (b) Hydrometeor type for scan in (a). (c) Hydrometeor type for PPI at 4° elevation angle. (d) DPR Ku-band reflectivity at 4 km. (e) Hail flag for scan in (d).

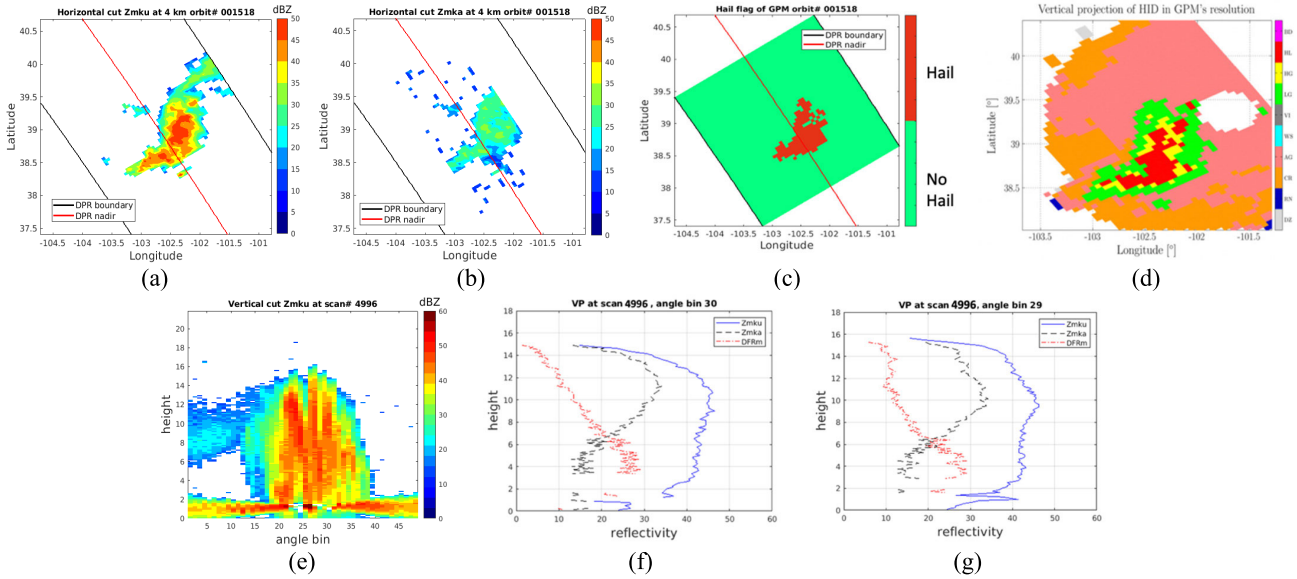


Fig. 9. Precipitation event simultaneously observed by KGLD radar on the border of Kansas and Colorado and GPM DPR orbit #1518 on 06-05-2014. (a) DPR Ku-band reflectivity at 4 km. (b) DPR Ka-band reflectivity at 4 km. (c) Hail flag for scan in (a) and (b). (d) Projection of HID for KGLD radar using GPM resolution (this plot is from [24, Fig. 3]). (e) DPR vertical cut of Ku reflectivity at scan #4996. (f) and (g) Vertical profile of Zmku, Zmka, and DFRm at bin #30 and #29 for scan #4996, respectively.

Unit-B (AMSU-B)/microwave humidity sounder (MHS)-based radiometers but has been fit to the observations of all microwave radiometers onboard the satellites of the GPM.

A hail case is studied in [26] near Veneto region in Italy on 2018-07-05. The hail detection is made using a microwave cloud classification method for hail detection (MWCC-H) model. This probability-based model originally designed for AMSU-B/MHS-based radiometers has been fit to the observations of all microwave radiometers onboard the satellites of the GPMs constellation. Fig. 10(a) shows the hail probability for this storm (this plot is taken from [26]). There are three regions showing hail in red circles. GPM DPR (orbit #24721) captured the same storm on 2018-07-05 around UTC 17:13:35. Fig. 10(b) and (c) shows DPR reflectivity at Ku- and Ka-bands at 4 km height. Red and black lines in Fig. 10(b) and (c) are DPR nadir and boundaries of the swath (version 7 data with full swath). The blue lines are the location of scan #4813 and #4801, where vertical cuts of reflectivity are shown in Fig. 10(d) and (e). Strong reflectivity is observed beyond 50 dBZ in vertical cuts of Fig. 10(d) and (e) with storm top reaching as high as 10–14 km. Fig. 10(f) shows the hail flag for the scan in Fig. 10(b). As expected, hail is detected with red color. The locations of hail in Fig. 10(f) align well with the locations at three red circles in Fig. 10(a). More

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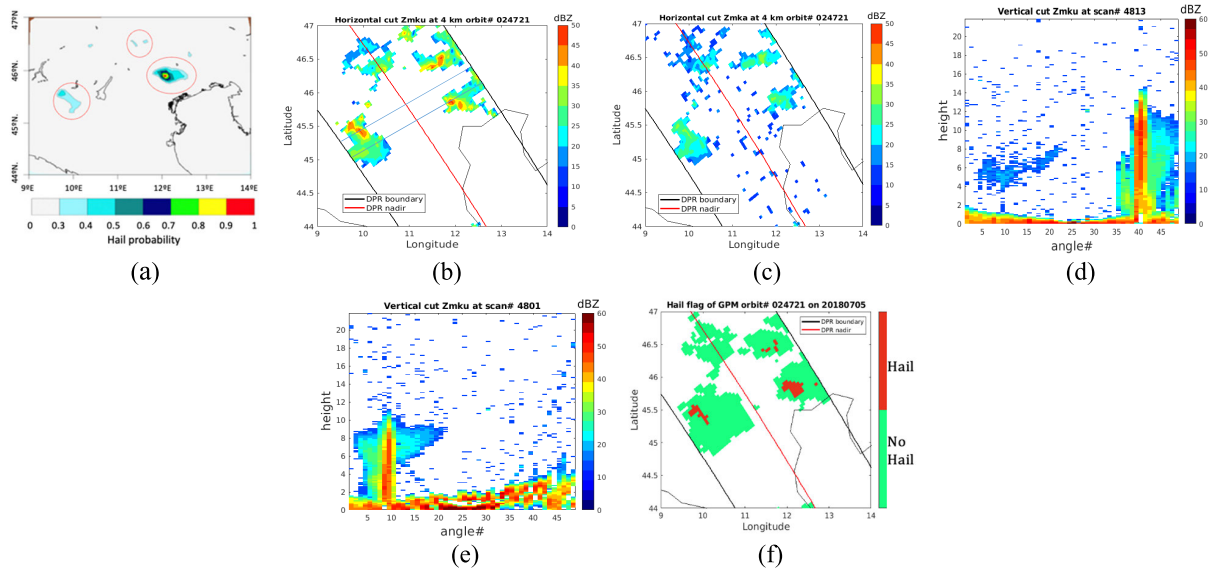


Fig. 10. Hailstorm detected by DPR (orbit #24721) on 2018-07-05 near Veneto region in Italy. (a) Hail detected with MWCC-H model (plot from [26]). (b) DPR Ku-band reflectivity at 4 km. (c) DPR Ka-band reflectivity at 4 km. (d) Vertical cut of DPR Ku-band reflectivity at scan #4813. (e) Vertical cut of DPR Ku-band reflectivity at scan #4801. (f) Hail flag for case in (b).

comparisons with GMI-based approach on a global scale are illustrated in Section V.

In [25], there is a list of "Champion hail cases" detected with GMI onboard the GPM. Within the list, we demonstrate two cases here showing the performance of our hail detection algorithm.

Fig. 11 shows a storm that happened east of Sydney, NSW, Australia, on 10/31/2020 observed by GPM DPR orbit #37918. The curved line in Fig. 11(a) marks the coastline of the city. Fig. 11(a) and (b) shows reflectivity from Ku- and Ka-bands of DPR radar at 4 km height. Strong reflectivity values around 50 dBZ can be observed. The vertical cross section at scan of 4995 is shown in Fig. 11(c). The height of the storm cell reaches about 16 km. In Fig. 11(d) and (e), two vertical profiles are chosen at angle bin 23 and 24 marked as two vertical dashed lines in Fig. 11(c). Blue, black, and red curves indicate Zmku, Zmka, and DFRm profile, respectively. Profiles at Fig. 11(d) and (e) are typical profile embedding hail features described in Section II with clear DFRm knee. Our hail flag is shown in Fig. 11(f) where the red color region successfully identifies hail.

Another "Champion hail cases" in [25] is from northern Argentina in South America. GPM DPR observed this hailstorm with orbit #33877 on 2020-02-14. Fig. 12(a) and (b) shows reflectivity from DPR at Ku- and Ka-bands at 4 km height. Enhanced reflectivity values reach 45–50 dBZ. The hail flag in Fig. 12(c) captured this hail precipitation successfully.

C. Validation With Hail Report and C-Band Radar

An exceptional hailstorm was observed over the Gulf of Naples in Italy 2015-09-05 and was recorded in [22]. The ground hail report and new image showed that the storm dropped 6–10 cm diameter hailstones along its path over the sea, and in Pozzuoli, near Naples. GPM radar overpass #8630 captured the storm in proximity of Naples at 8:47 UTC.

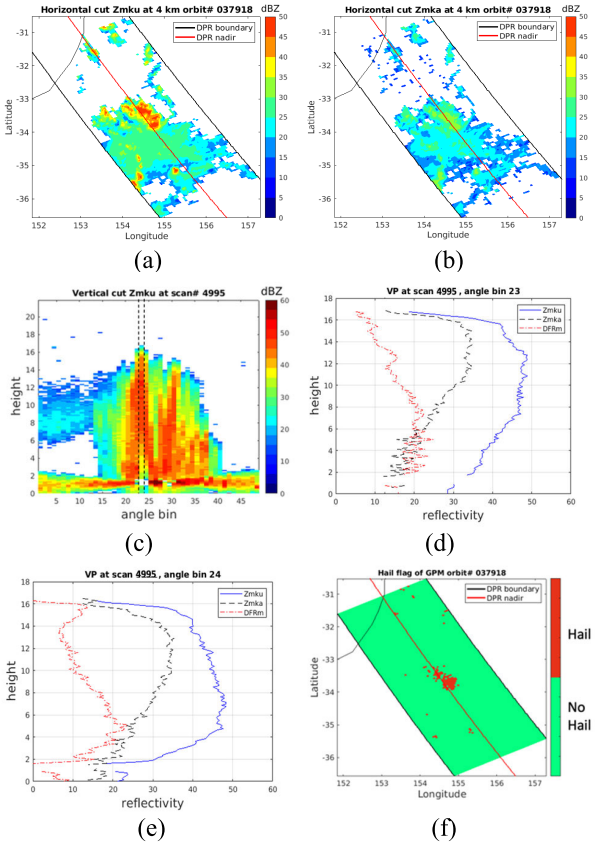


Fig. 11. Hailstorm east of Sydney, Australia, on 2020-10-31 captured by GPM orbit #37918. The storm is listed as one of the "Champion hail cases" in [25]. (a) Zmku at 4 km from DPR observation. (b) Zmka at 4 km from DPR observation. (c) Vertical cross section of Zmku at scan #4995. (d) and (e) Vertical profile of Zmku, Zmka, and DFRm at angle bin #23 and #24 from scan #4995. The locations are indicated as two vertical dashed lines in (c). (f) Hail flag for the case.

A sequence of reflectivity factor maps collected by the Monte il Monte radar (C-band polarimetric) from 08:30 UTC to 10:55

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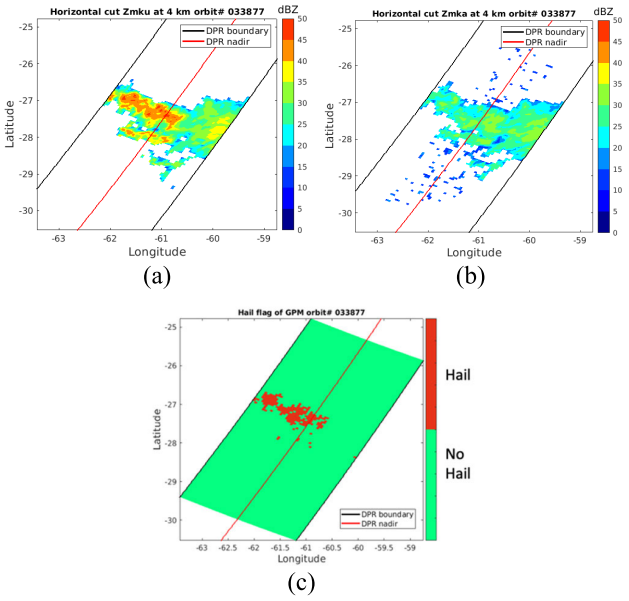


Fig. 12. Hailstorm in northern Argentina on 2020-02-14 observed by GPM with orbit #33877. This storm is listed as one of the "Champion hail cases" in [25]. (a) Zmku at 4 km from DPR observation. (b) Zmka at 4 km from DPR observation. (c) Corresponding hail flag of this case.

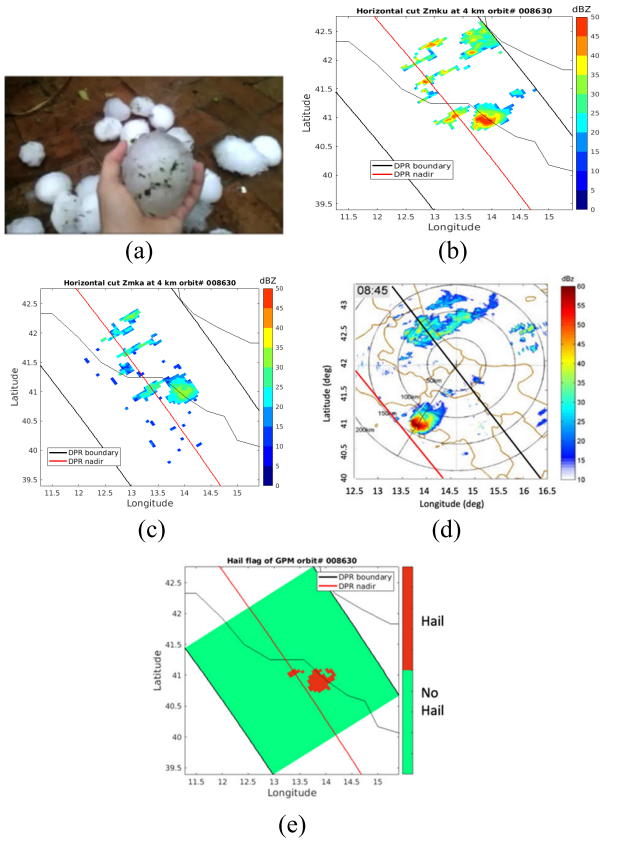


Fig. 13. Hailstorm observed over the Gulf of Naples in Italy on 2015-09-05 (see [22]). GPM overpass captured the storm in proximity of Naples at 8:47 UTC with orbit #8630. (a) Hail report by a boat showing the storm dropped 6–10-cm-diameter hailstones along its path over the sea, and in Pozzuoli, near Naples (from [30, Fig. 1]). (b) DPR observation of Zmku at 4 km. (c) DPR observation of Zmka at 4 km. (d) Reflectivity factor maps collected by the Monte il Monte radar (C-band polarimetric) at 08:45 UTC (from [30, Fig. 6]). (e) Hail flag of this hailstorm.

UTC highlighting the evolution of the hailstorm. Fig. 13(a) shows the actual hail size compared to an adult's palm size. Fig. 13(b) and (c) shows the Ku- and Ka-band reflectivity from DPR radar at 4 km. Peak reflectivity for Ku-band is around 50 dBZ. In Fig. 13(c), only inner swath of Ka-band data is available since the storm was observed before DPR scan pattern change in May 2018. Fig. 13(d) shows the Monte il Monte radar data at UTC 8:45. Extreme high value beyond 60 dBZ is observed near DPR nadir direction (red solid line). In Fig. 13(e), the hail flag is illustrated, and the red color indicates hail precipitation. The location is well aligned with the C-band peak reflectivity region.

D. Validation With Multiple Scattering Effect in Level-2 Algorithm

In [24], a hail validation case is studied between KFWS radar and Trigger module outputs from the GPM DPR level-2 algorithm. In this section, we will study the same validation case but adding the performance of "flagHail" from the hail identification algorithm. A new module called "Trigger module" has been implemented in the DPR level-2 algorithm version 7 [18]. The Trigger module aims at detecting the occurrence of severe multiple scattering effect and nonuniform beam filling within each DPR footprint. There is an index in the Trigger module summarizing the presence of multiple scattering (called "MS index") where 0 represents absent, 100 represents certain, and 50 means possibly present/uncertain. Under most circumstances, strong hail profiles illustrate multiple scattering properties, while profiles with MS effects do not necessarily mean that it is hail. However, there is strong indication between these two. Typical multiple scattering effects include the presence of "DFR knee" along the profile, absence of reflectivity peak at surface, anomalous Ka reflectivity profile compared to that of the Ku channel, and so on. Details of multiple scattering features can be found in [25].

Fig. 14 shows the detailed comparisons between NEXRAD radar HID, hail flag from hail identificaiton algorithm, and MS index from Trigger module for the hailstorm observed by the GPM DPR (orbit #6906) on 11-05-2015 in the Dallas area together with the nearest NEXRAD radar of KFWS. Fig. 14(a) shows a PPI scan of KFWS S-band reflectivity at the elevation of 2.4°. The black thick line and red line are the DPR boundary and nadir direction. Strong reflectivity can reach around 50–55 dBZ. Three major regions with the highest reflectivity are marked with black circles. In Fig. 14(b), hydrometeor identification (HID) is illustrated using the algorithm from [29]. The red color in Fig. 14(b) indicates hail, as circled with black color. Horizontal reflectivity structures from GPM DPR at 4 km height are shown in Fig. 14(c) and (d). Even with strong attenuations from hailstorm, peak reflectivities are still around 45–50 dBZ for Ku-band and 30–35 dBZ for Ka-band. Fig. 14(e) shows the hail flag from this hail identification algorithm using observations from Fig. 14(c) and (d). Red color is for DPR profile with hail along the profile. Although with much rough resolution from space radar, the hail algorithm successfully identifies hail at the same locations as

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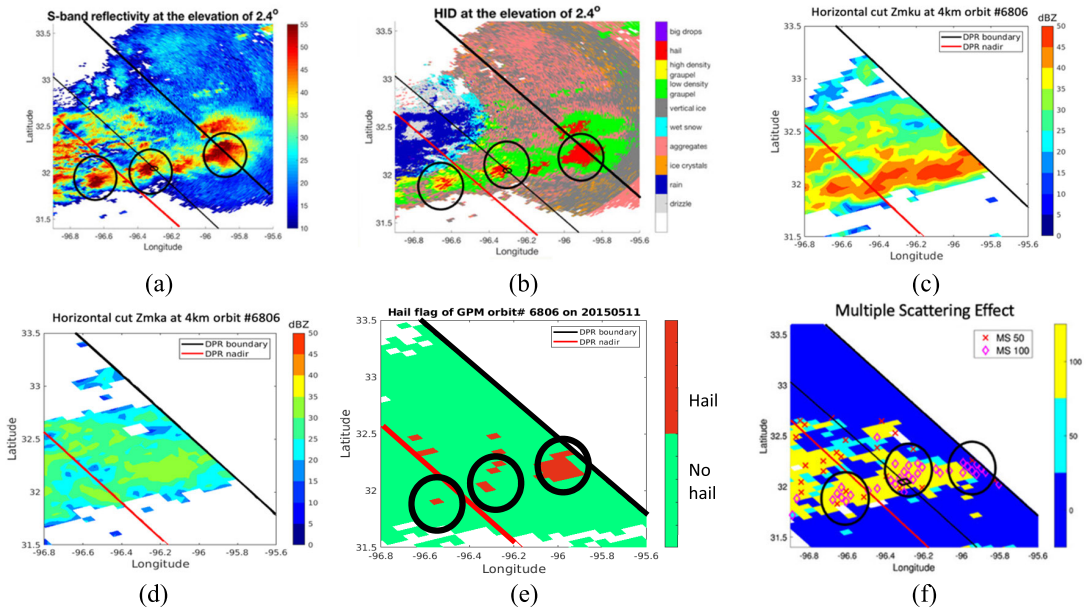


Fig. 14. Thunderstorm observed by the GPM DPR (orbit #6906) on 11 May 2015 in the Dallas area together with the nearest NEXRAD radar (KFWS) at the elevation of 2.4°. (a) HID field corresponding to the measurements from the cut. Thick black and red lines are DPR boundary and nadir, respectively. (a) and (b) Plots taken from [23]. (c) DPR Ku-band reflectivity at 4 km. (d) DPR Ka-band reflectivity at 4 km. (e) Hail flag for case in (c). (f) Multiple scattering effect index from Trigger module of DPR level-2 algorithm. MS index shown by different markers is described in the legend. "MS 100" means multiple scattering certain. (f) Plot taken from [24].

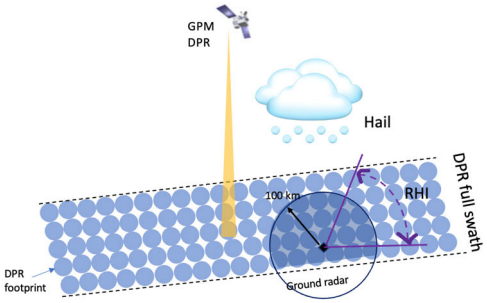


Fig. 15. Cartoon illustration to depict the approach performing quantitative validation between hail identification algorithm and ground radar hydrometeor types.

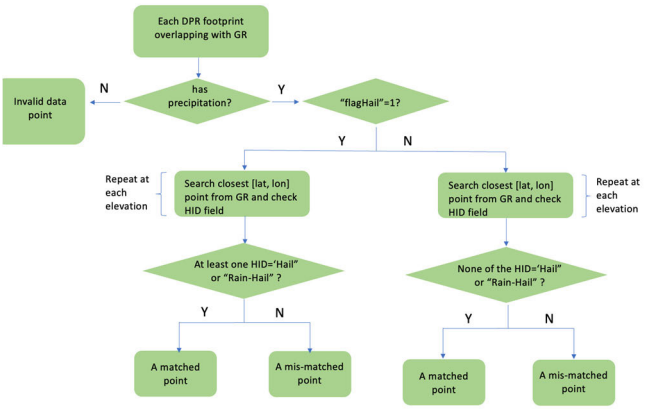


Fig. 16. Flowchart to perform match ratio calculation between DPR and ground radar.

from ground radar. The output of the Trigger module for the corresponding case is shown in Fig. 14(f), and the rhombus shapes depict "MS 100," which means multiple scattering certain. Most "MS 100" cases are within the three black circles matching the findings in Fig. 14(a), (b), and (e). This is a clear indication that strong correlation exists between hail profiles and presence of multiple scattering effects. The correlation between hail and multiple scattering is an interesting topic, but the content is beyond the topic of this article.

E. Quantitative Analysis

Besides qualitative comparison, we further enhance the evaluations with quantitative analysis. Fig. 15 shows a cartoon illustration that shows how quantitative analysis is performed in this study.

Solid blue circles indicate the footprints for DPR full swath where they are the matched Ku- and Ka-band observations. The big blue circle is the 100-km-range circle for ground radar at the lowest elevation angle (around 0.58°). Shaded areas are the overlap region between DPR and ground radar. The match ratio between DPR and ground radar is calculated within this area. Whenever the DPR footprint has a valid 0 or 1 value from the "flagHail," we search the ground radar hydrometeor type at the closest latitude and longitude data point. When the hail flag is 1, we consider hydrometeor types of "Hail" and "Rain-Hail" (see [28]) from ground radar to be a match. We repeat this search for each elevation angle for ground radar until we find a match. If there is no match for the whole vertical volume, then we consider that this DPR footprint does not have a matched hydrometeor type from ground radar. When "flagHail" is 0 at the DPR footprint, the other hydrometeor types (other than "Hail" or "Rain-Hail") from the ground radar identification algorithm are considered a match. Similarly, we will repeat at each elevation angle from ground radar. If there is no "Hail" or "Rain-Hail" at each elevation angle, we consider this a matched hydrometeor type for the comparison. Fig. 16 shows the flowchart for the match process. The equation we use to

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TABLE III
MATCH RATIO FOR FOUR VALIDATION CASES BETWEEN DPR AND NEXRAD RADAR

	KLZK VS DPR orbit 25427	KLNX VS DPR orbit 25028	KFSD VS DPR orbit 24198	KVNX VS DPR orbit 24557	KGLD VS DPR orbit 1518	KFWS VS DPR orbit 519
Location and Time	Little rock, AR 8/20/2018	North Platte, NE 7/25/2018	Sioux Falls, SD 6/2/2018	Vance AFB, OK 6/25/2018	Boarder of CO and KS, 6/5/2014	Dallas Fort Worth, TX, 4/2/2014
Valid data point	606	741	1054	581	788	623
Matched data point	532	731	1019	545	745	577
Match ratio	**87.8%**	**98.6%**	**96.7%**	**93.8%**	**94.5%**	**92.6%**

calculate the match ratio is described in the following:

$$\text{Match ratio} = \frac{\#\text{ of matched DPR footprint}}{\#\text{ of valid DPR footprint overlapping with GR}} \tag{8}$$

valid footprints in the denominator of (8) mean that both DPR and ground radar detect precipitation. If the match ratio equals 1, it means that all DPR footprints can find a match of Hail with ground radar. It needs to be noted that the match ratio is calculated when both DPR and ground radar have precipitation. DPR has limited detectability due to its attenuation at higher frequencies compared to the ground radar. It is normal that valid DPR points are less than the ground radar.

Six comparison cases are carefully chosen from seven states of continental USA (CONUS) where hail is most popular, including Kansas, Colorado, Arkansas, Nebraska, South Dakota, Oklahoma, and Texas. One of the validation radars KGLD is located on the border of Colorado and Kansas. Table III lists the details of the events and the results of match ratio for these six validation cases. From the table, the highest match ratio reaches 98.6% and the lowest is 87.8% with the average equaling 94% among these six cases. This preliminary analysis gives out satisfactory match ratio between two radar systems. In the future study, we will perform more quantitative analysis based on a large scale.

V. GLOBAL DISTRIBUTION OF HAIL AND CROSS COMPARISON

A. Hail Distribution From Product of "flagHail"

Spaceborne radar, such as GPM, offers a unique 3-D view of the atmosphere over the oceans and land difficult to access in remote areas. One of the major goals for spaceborne radar is to study the global distribution of precipitation and the trend in the long term. Fig. 17(a) shows a global map with "flagHail" counts accumulated in the 2° × 2° latitude/longitude box using the full swath data of three years from 2019 to 2021. In each grid box, "flagHail" (=1 when hail exists along the profile) value is added accumulatively. The percentage is calculated based on the total number of "flagHail" counts. It needs to be mentioned that the "flagHail" count is not equivalent to the number of hailstorms but is a good indication of hail distribution.

The map clearly shows the importance of orography and land–sea contrast in hail formation. According to the map, most of the hail events occur over land and over regions adjacent to the land. There are eight red squares indicating the regions with high hail frequency. The regions are Africa region

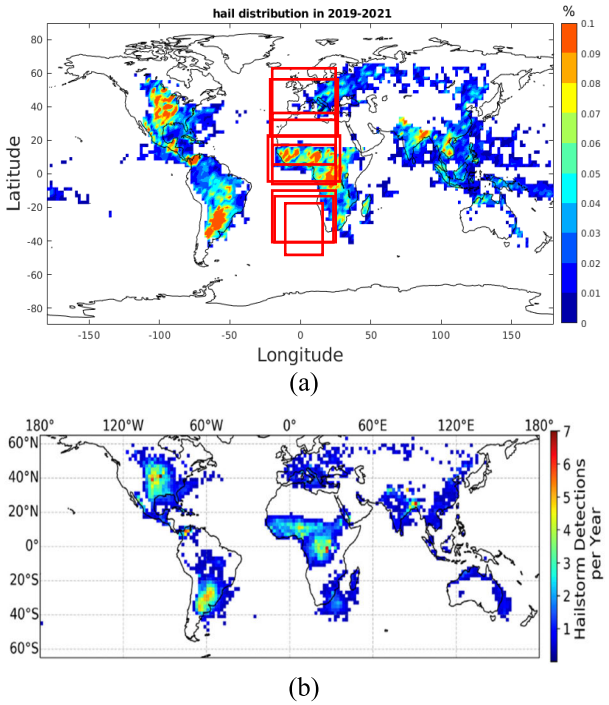


Fig. 17. (a) Global distribution of "flagHail" counts on a 2° × 2° lat/lon grid over globe using GPM DPR full swath data from year 2019 to 2021. (b) Same figure from [25, Fig. 9.10]. Hail climatology based on the approach of Bang and Cecil [27], as applied to both TMI and GMI.

south of Sahel and north of Kalahari Desert; southern Africa, including the island of Madagascar; the subtropical Americas, including southern Brazil, northern Argentina, and Central United States; foothill of Himalaya region; and northern and eastern Australia. Fewer counts are scattered in northern Asia and extend to southeast of Europe. For simplicity, we use abbreviation of NAM, NEU, SAS, SAF, AUS, SAM, CAF, and CAM to represent these high hail frequency regions in North America, North Europe, South Asia, Southern Africa, Australia, South America, Central Africa, and Central America.

Other existing hail detection algorithms from space that have been developed either use deterministic thresholds selected for satellite measurements or were formulated on probabilistic approaches (see [8], [9], [24], [25], [26], [27]). Fig. 17(b) shows an example of this approach. This is the same figure from [25, Fig. 9.10] and it is here for convenience of comparison. The plot uses the probabilistic approaches that yield a weighted count, with weights proportional to the probabilities from individual storms. Our algorithm uses only DPR radar parameters but with a statistically based PTI threshold.

Comparing Fig. 17(a) and (b), virtually, these two satellite-based hail detections agree on the eight regions marked by red squares in Fig. 17(a). They are located in northern Argentina, the central United States, both eastern India/Bangladesh and western India/Pakistan, central Africa, and north/east of Australia. There are some disagreements on the relative magnitudes of those hot spots and some other regions as small scatters of hail in northern Asia and eastern Europe in the summer season. We also detect a small number of hailstorms

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over warm waters in west pacific close to continental regions. Besides that, the level of agreement between these two methods is remarkable considering the huge difference between radar- and radiometer-based algorithms. More ground validation activities are needed for further validations.

Monthly "flagHail" counts are shown in Fig. 18(a)–(c). In Fig. 18(a), obvious hail counts increase from May to September for regions of NAM, NEU, and SAS, with the NAM having the most growth. These months correspond to the spring and summer season for the north hemisphere. In Fig. 18(b), on the other hand, regions in SAF, AUS, and SAM show opposite monthly trend of "flagHail" counts where more hails are detected from October to March. For regions near Equator such as CAF and CAM, hail counts are relatively steady among the months, as shown in Fig. 18(c). Fig. 18(d) shows a pie chart of hail distribution among eight regions with three-year hail data. Regions, such as SAS, NAM, CAF, and SAM, are top four major areas that have the most hail precipitation.

We observe severe hailstorms in North America that extend close to 55°N. Almost no hailstorms are detected south of 40°S. Around 86.3% of satellite-inferred hail precipitation is occurred within 45° latitudes. Fig. 18(e) shows this zonal dependence of hail distribution. These findings are consistent with those in [9], [24], and [30].

B. Quantitative Comparison With Other Radar-Based Hail Identification Algorithm

In [24], proxies for identification of hail (including aloft and on surface) were developed from GPM observations. The proxies were tested using the critical success index (CSI) as a verification measure and found out that the algorithm developed based on the combination of mean reflectivities at Ku- and Ka-bands in the mixed-phase layer performs the best (CSI = 48.7%) when validating with ground truth. The mixed-phrase reflectivity is defined as the average over the 4-km layer of the measured reflectivity above −10 °C isotherm using the equation in (9) [24, eq. (8)]. In the comparison, we use a temperature lapse rate of 6.5 k/km to decide the number of bins used in the calculation

$$Z^{\text{mix}} = 10\log_{10}\left[\int_{-10^\circ C}^{-10^\circ C+4km} Z(h)dh/4km\right]. \tag{9}$$

The criterion for the optimal proxy for hail detection using the mixed phase reflectivity at both DPR frequencies is listed as follows [24, eq. (11)]:

$$\begin{cases} Z_{\text{Ku}}^{\text{mix}} > 0.632 Z_{\text{Ka}}^{\text{mix}} + 20.4 \\ Z_{\text{Ku}}^{\text{mix}} > 40.15. \end{cases} \tag{10}$$

In this section, for comparison purposes, we set this hail flag from [24] that equals 1 when mixed-phrase reflectivity meets the above condition at each valid DPR footprint. Next, we accumulate the hail counts in 2° × 2° latitude/longitude grid and normalized by the total hail counts. Fig. 16 shows this hail distribution using full-year data from 2021. Fig. 19(a) shows the hail distribution using Mroz et al. [24] method and Fig. 19(b) shows the distribution using our algorithm.

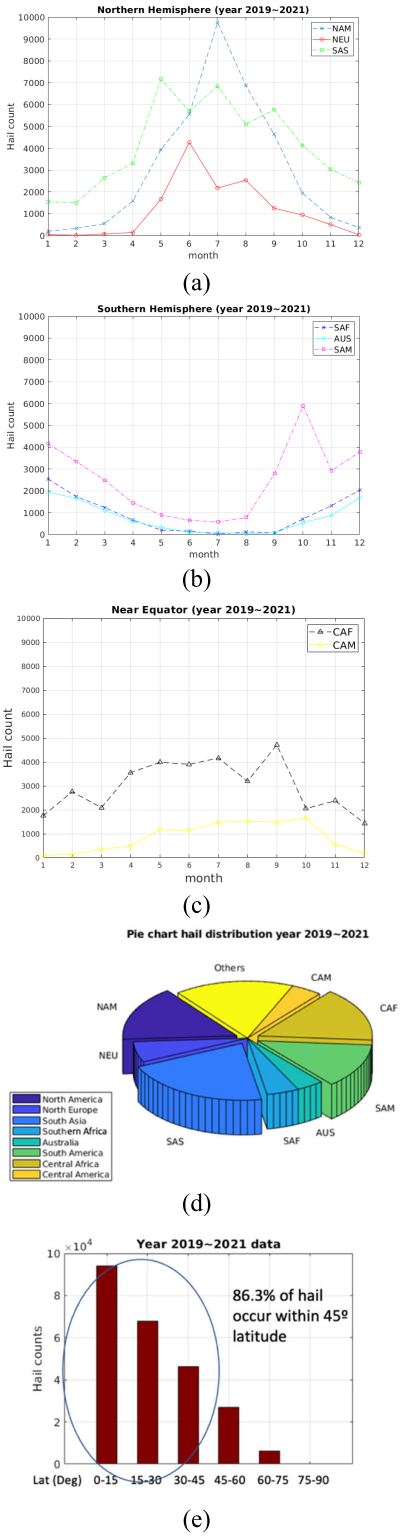


Fig. 18. Statistics of "flagHail" counts using three-year full swath data from 2019 to 2021. (a) Monthly trend of "flagHail" counts for regions in the Northern Hemisphere. (b) Monthly trend of "flagHail" counts for regions in the Southern Hemisphere. (c) Monthly trend of "flagHail" counts for regions near the Equator. (d) Pie chart of hail distribution for eight regions. (e) Zonal distribution of "flagHail" counts.

From virtual comparison, hail counts peak at eight regions marked as red squares in Fig. 17 from both algorithms. Our algorithm detects more hail in the central America and south

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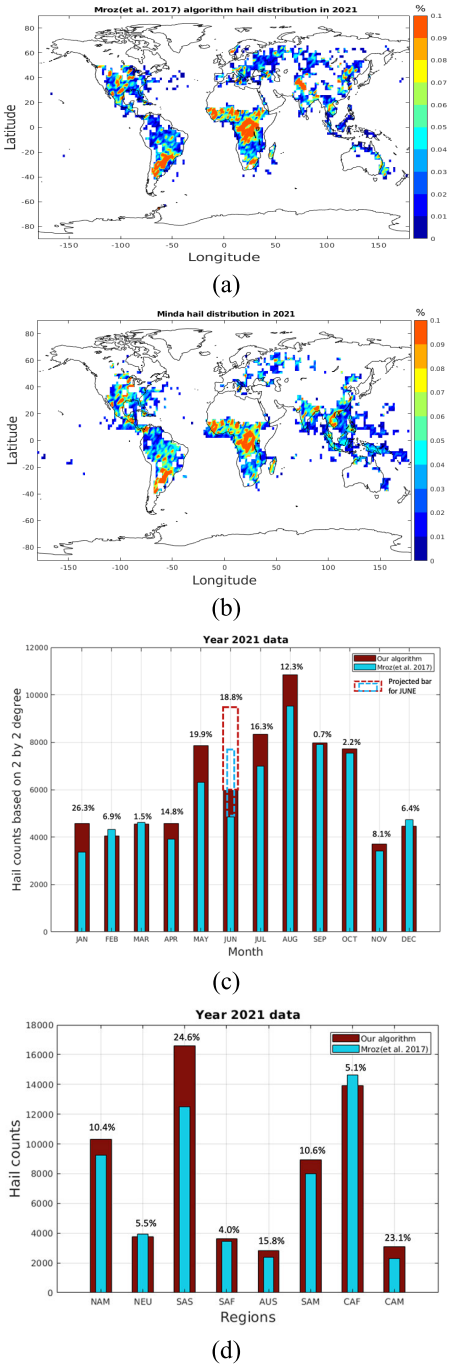


Fig. 19. (a) Global distribution of hail counts on a 2° × 2° lat/lon grid using Mroz et al. [24] algorithm using data from year 2021. (b) Same as (a) with our hail detection method ("flagHail"). (c) Quantitative comparison of hail counts on a monthly base. (d) Quantitative comparison of hail counts on a regional base.

Asia and Mroz et al. [24] method detects more hail in the northern Europe. Quantitatively, Fig. 19(c) shows the bar plots showing the percentage of mismatch for each month between these two algorithms. The percentage is calculated using the absolute difference normalized by the total hail count. Both bars have the most hail counts in month from May to October. Month of June has a dip in the counts due to the GPM data missing from 6/18 to 6/30 on the NASA GPM data site (https://disc.gsfc.nasa.gov/datasets/GPM_2ADPR_07). The dashed lines are a projection of the full month data. The mismatch ratio ranges from 0.7% in September to 26.3% in January. The average ratio is 11.2% based on 12 months. Regionwise, the bar plot in Fig. 19(d) lists the hail counts in eight regions from both algorithms. The biggest difference in hail detection happens in the south Asia with the mismatch ratio of 24.6%. This difference can also be spotted when comparing Fig. 19(a) and (b). Our algorithm detects more hail in the intertropical convergence zone (ITCZ) region and tropical islands. Other seven regions show a relatively small mismatch ratio. The average of mismatch ratio for all eight regions is less than 13%. These are encouraging results for a global-based hail comparison.

VI. Summary and Conclusion

The aim of this work is to introduce a new hail identification algorithm for GPM-DPR radar. This algorithm has been implemented in the GPM DPR level-2 algorithm of version 7 since December 2021. The product, for the first time, offers the potential of retrieving a uniform and homogeneous hail dataset on the global scale from dual-frequency space radar sensors. The new product of "flagHail" in the classification module identifies whether hail exists (=1) or not (=0) along each DPR vertical profile, but not necessarily on the ground. This hail identification algorithm is built upon PTI, which is defined by the dual-frequency characteristics along the Ku- and Ka-profiles of DPR. PTI has been widely applied to authors' previous research work developing algorithms for surface snowfall and GH identification and proved to be very efficient and useful.

The hail identification algorithm is cross-compared with various sources, including hail ground report, ground radar system of NEXRAD, radiometer-based algorithm, and algorithm developed by the Trigger module in the GPM DPR level-2 algorithm. Validations illustrate generally good performance on hailstorm cases. Quantitative analysis is performed with NEXRAD radar cases and illustrates around 94% of match ratio. Besides specific cases, the hail algorithm needs to be validated on a bigger scale. Three years from 2019 to 2021, DPR data are used to generate a global distribution map of hail occurrence. It needs to be noticed that the counts of "flagHail" are applied in the plot on a 2° × 2° latitude/longitude degree resolution. The map clearly shows the importance of orography and land–sea contrast in hail formation. Most of the hail events occur over land and over regions adjacent to the land. The regions having local maxima of hail are Africa region south of Sahel and north of Kalahari Desert, the subtropical Americas including southern Brazil, northern Argentina and Central United States, foothill of Himalaya region, and northern Australia. Fewer counts are scattered in northern Asia and extend to southeast of Europe. Hailstorms can be found in Central Africa throughout the year. The seasonal transition of hail distribution is clear in the monthly hail counts. Obvious hail counts increase from May to September for regions of South Asia and regions of Europe and USA, with the United States having the most growth. On the other hand, regions in South Hemisphere, such as Australia and South American regions, show the opposite monthly trend of "flagHail" counts.

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This global map is cross-compared virtually with the hail map generated with GMI-based approach using brightness temperature models and statistical models. Common hot spots of hail occurrence are spotted among various algorithms. Besides the common findings, our algorithm detect small scatters of hail in northern Asia and eastern Europe in the summer season and detect small amount of hailstorms over warm waters in west pacific close to continental regions. More ground validation activities are needed for further validation in these areas. Considering the huge difference between radar- and radiometer-based algorithm, the hail detection from both sources demonstrate similar global hail distribution in general.

Quantitatively, we compare our method with Mroz et al.'s [24] criteria using mixed-phrase reflectivity from dual-frequency observations. Based on a year of data (year 2021), the mismatch ratio for monthly and regional comparison is all below 13%. These are encouraging results on a global-scale comparison.

Ongoing work is focused on providing more quantitative validations in the ITCZ region either through hailstorm cases or large-scale statistics and polishing the hail identification algorithm under various climatological and geographical conditions.

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